

SECSR2043 OPERATING SYSTEMS
[20 Marks]

Name : AIN NURNABILA BINTI MOHD AZHAR Student : A23S0207 ID : Section : 2 	Marks
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Instruction: Please answer all of the following questions. Whenever the 🖐 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: ahmad.sh).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.sh .; cp ../fruits/*.sh .;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]

```

bella@secr2043:~$ nano bella.sh
bella@secr2043:~$ chmod u+x bella.sh
bella@secr2043:~$ ./bella.sh
bella@secr2043:~$ ls -R
.:
bella.sh cars dates drinks fruits helloworld.jar

./cars:
accord.c civic.c proton.c

./dates:
appointment dateoftheday

./drinks:
accord.c apple.txt civic.c helloworld.jar orange.txt proton.c

./fruits:
apple.txt orange.txt

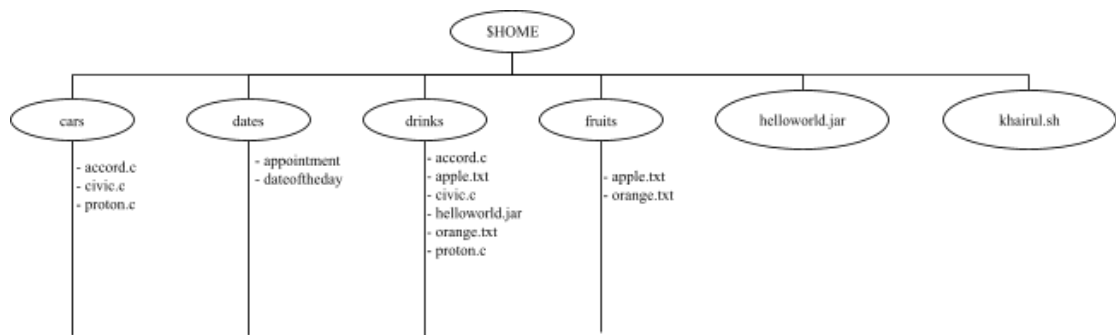
```



```

#!/bin/bash
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd ../drinks; cp ../cars/*.c .; cp ../fruits/*.txt .;
cp ../*.jar .

```



- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter “c” as the input to the bash script. [4 marks]



```
#!/bin/bash

echo "Enter the file extension:"
read ext

echo "Files with extension .$ext:"
find . -type f -name ".*$ext"

count=$(find . -type f -name ".*$ext" | wc -l)
echo "Total number of .$ext files: $count"
```

```
bella@secr2043:~$ nano count.sh
bella@secr2043:~$ chmod u+x count.sh
bella@secr2043:~$ ./count.sh
Enter the file extension:
txt
Files with extension .txt:
./fruits/apple.txt
./fruits/orange.txt
./drinks/apple.txt
./drinks/orange.txt
Total number of .txt files: 4
```



2. The following Figure 1 illustrates a tree structure of some directories and files.

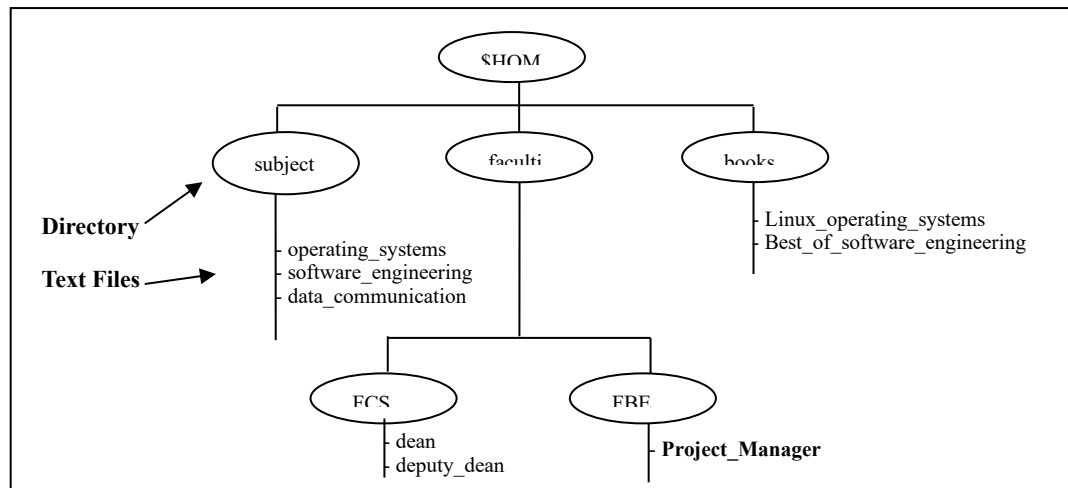


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

```

bella@secr2043:~$ nano bella2a.sh
bella@secr2043:~$ chmod u+x bella2a.sh
bella@secr2043:~$ ./bella2a.sh
bella@secr2043:~$ ls -R
  
```

```
bella.sh bella2a.sh books cars count.sh dates drinks faculties fruits helloworld.jar subjects
./books:
Best_of_Software_Engineering Linux_Operating_Systems
./cars:
accord.c civic.c proton.c
./dates:
appointment dateoftheday
./drinks:
accord.c apple.txt civic.c helloworld.jar orange.txt proton.c
./faculties:
FBE FCS
./faculties/FBE:
Project_Manager
./faculties/FCS:
dean deputy_dean
./fruits:
apple.txt orange.txt
./subjects:
data_communication operating_systems software_engineering
```



- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called `book`.

[2 marks]

```

bella@secr2043:~$ ls -l
total 44
-rwxr--r-- 1 bella bella 382 Jun 30 23:53 bella.sh
-rwxr--r-- 1 bella bella 530 Jul 1 00:02 bella2a.sh
drwxr-xr-x 2 bella bella 4096 Jul 1 00:03 books
drwxr-xr-x 2 bella bella 4096 Jun 30 23:51 cars
-rwxr--r-- 1 bella bella 214 Jun 30 23:54 count.sh
drwxr-xr-x 2 bella bella 4096 Jun 30 23:51 dates
drwxr-xr-x 2 bella bella 4096 Jun 30 23:51 drinks
drwxr-xr-x 4 bella bella 4096 Jul 1 00:03 faculties
drwxr-xr-x 2 bella bella 4096 Jun 30 23:51 fruits
-rw-r--r-- 1 bella bella 12 Jun 30 23:51 helloworld.jar
drwxr-xr-x 2 bella bella 4096 Jul 1 00:03 subjects
bella@secr2043:~$ ls -l books
total 8
-rw-r--r-- 1 bella bella 29 Jul 1 00:03 Best_of_Software_Engineering
-rw-r--r-- 1 bella bella 24 Jul 1 00:03 Linux_Operating_Systems
bella@secr2043:~$ ls -l faculties/FCS
total 8
-rw-r--r-- 1 bella bella 5 Jul 1 00:03 dean
-rw-r--r-- 1 bella bella 12 Jul 1 00:03 deputy_dean
bella@secr2043:~$ ls -l faculties/FBE
total 4
-rw-r--r-- 1 bella bella 16 Jul 1 00:03 Project_Manager
bella@secr2043:~$ ls -l faculties
total 8
drwxr-xr-x 2 bella bella 4096 Jul 1 00:03 FBE
drwxr-xr-x 2 bella bella 4096 Jul 1 00:03 FCS

```

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxr-xr-x
Best_of_software_engineering	-rw-r--r--
FCS	drwxr-xr-x
project_manager	-rw-r--r--



- c) Write another bash script (called myname2c.sh) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users		
	Owner	Group	Public

subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

```
#!/bin/bash
```

```
chmod 744 subjects
chmod 520 books/Best_of_Software_Engineering
chmod 607 faculties/FCS
chmod 031 faculties/FBE/Project_Manager
```

```
bella@secr2043:~$ nano bella2c.sh
bella@secr2043:~$ chmod u+x bella2c.sh
bella@secr2043:~$ ./bella2c.sh
bella@secr2043:~$ ls -l
total 48
-rwxr--r-- 1 bella bella 382 Jun 30 23:53 bella.sh
-rwxr--r-- 1 bella bella 530 Jul 1 00:02 bella2a.sh
-rwxr--r-- 1 bella bella 141 Jul 1 00:16 bella2c.sh
drwxr-xr-x 2 bella bella 4096 Jul 1 00:03 books
drwxr-xr-x 2 bella bella 4096 Jun 30 23:51 cars
-rwxr--r-- 1 bella bella 214 Jun 30 23:54 count.sh
drwxr-xr-x 2 bella bella 4096 Jun 30 23:51 dates
drwxr-xr-x 2 bella bella 4096 Jun 30 23:51 drinks
drwxr-xr-x 4 bella bella 4096 Jul 1 00:03 faculties
drwxr-xr-x 2 bella bella 4096 Jun 30 23:51 fruits
-rw-r--r-- 1 bella bella 12 Jun 30 23:51 helloworld.jar
drwxr--r-- 2 bella bella 4096 Jul 1 00:03 subjects
bella@secr2043:~$ ls -l books
total 8
-r-x-w---- 1 bella bella 29 Jul 1 00:03 Best_of_Software_Engineering
-rw-r--r-- 1 bella bella 24 Jul 1 00:03 Linux_Operating_Systems
bella@secr2043:~$ ls -l faculties
total 8
drwxr-xr-x 2 bella bella 4096 Jul 1 00:03 FBE
drw----rwx 2 bella bella 4096 Jul 1 00:03 FCS
bella@secr2043:~$ ls -l faculties/FBE
total 4
-----wx--x 1 bella bella 16 Jul 1 00:03 Project_Manager
```



- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr--r--
Best_of_software_engineering	-r-x-w----
FCS	drw----rwx
project_manager	-----wx--x

End of Lab 3

**** All the Best for Final Exam ****